Bridging Latent Traits and Epistemic States: Aligning Agent Reasoning under Dilemmas with Expert Cognition

Anonymous ACL submission

Abstract

Large Language Models (LLMs) have transcended text utilities to become generative mirrors of the human mind, capable of replicating complex cognition. While utilizing these mirrors to simulate expert personas remains a major pursuit, LLM-based agents relying on profiles fail to mimic authentic reasoning under dilemmas due to surface-level alignment. To bridge this gap, we treat professional design as a testbed for non-linear cognition, thereby introducing a framework that operationalizes Mischel's Cognitive-Affective Personality System theory as a cognitive engine. By mapping a 30-dimensional trait layer into a PKVC state generator, this framework leverages behavioral data from 162 designers to encode empirical human trajectories into ontological procedures, dynamically modulating state transitions to guide the agent's reasoning along expert decision paths under high-conflict scenarios. Evaluations show that our framework significantly outperforms standard persona baselines, paving a novel methodology to formally decrypt, simulate, and scrutinize the underlying architecture of human expertise.

1 Introduction

> *"Language is not merely a tool for communication, but a mirror of the mind."*
> —— *Noam Chomsky*

Simulating domain-specific expert personas has emerged as a fundamental pursuit for Large Language Model (LLM) based role-playing agents (Shanahan et al., 2023). However, when deployed to mimic authentic reasoning and decision-making under complex professional dilemmas, these agents consistently fail to produce high-fidelity behavioral trajectories (Du et al., 2026). The core bottleneck is not a deficiency of factual knowledge. While LLMs excel at describing the static attributes of a domain, they lack the underlying computational cognitive structure to mirror how a human expert actually reasons through a crisis. Authentic decision-making under high-conflict dilemmas is inherently nonlinear and dynamic, requiring a fluid shifting between redefining a problem, invoking past experiences, balancing competing values, and generating principles for action (Mischel and Shoda, 1995). To bridge this gap, advanced agents deploy alignment techniques to mirror stable character profiles (Shao et al., 2023). However, these methods rely on static declarative information or post-hoc constraints to dictate what an agent should express. They fail to model how an expert navigates the nonlinear, situated cognitive dynamics under conflict, nor can they externalize the rapidly shifting tacit knowledge embedded within professional choices. Consequently, reasoning chains (Zhang et al., 2025) collapse into flat, general-purpose logic, missing the domain-specific state transitions that define true human expertise. We term this limitation "surface-level persona alignment."

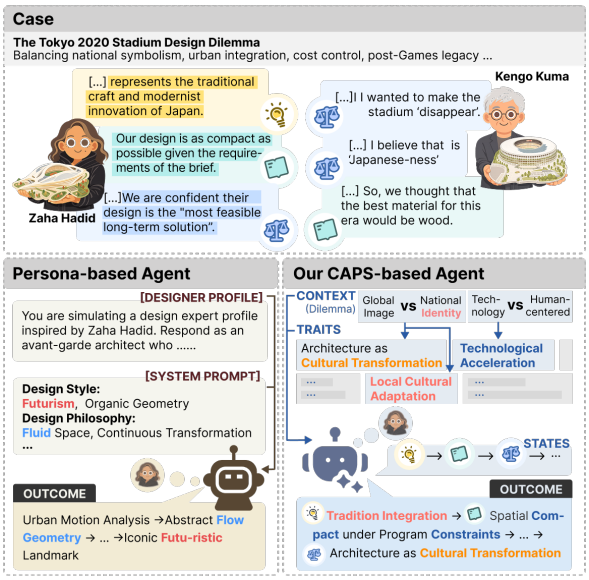


Figure 1: Example of the paradigm shift from surface-level persona alignment to deep cognitive alignment under the professional design dilemma.

Confronting these limitations, we pursue a paradigm shift toward **latent trait and epistemic state alignment** by treating professional design as an ideal testbed for non-linear cognition, as illustrated by the concrete design dilemma in Figure 1. Design reasoning represents a pinnacle of human problem-solving, where practitioners must continuously make high-stakes choices under wicked dilemmas. Under this testbed, we introduce a framework that operationalizes Walter Mischel's seminal **Cognitive-Affective Personality System (CAPS) theory (Mischel and Shoda, 1995) into a cognitive engine**. Our framework bridges latent traits of human experts with instantaneous reasoning states across two integrated layers: a 30-dimensional trait layer grounded in psychological constructs but localized to professional dilemmas, and a 4-state generator termed PKVC that formalizes the transitions of expert cognition: **Problem framing (P), Knowledge retrieval (K), Value judgment (V), and Concept generation (C)**.

The core innovation lies in the dynamic and computational interaction between these layers. To ground this interaction empirically, we leverage behavioral tracking data from 162 professional designers facing high-conflict dilemmas to discover and extract formal ontological procedures that explicitly capture how these 30-dimensional psychological traits systematically guide the shifting between P, K, V and C. At runtime, these ontological procedures dynamically parameterize and modulate the state-transition. Consequently, when the agent encounters a complex professional dilemma, its token generation is routed along authentic, non-linear expert paths, enforcing deep cognitive fidelity over surface-level imitation.

To our knowledge, this is the first framework to operationalize psychological CAPS theory to control the reasoning trajectories of LLM-based agents under complex dilemmas. In a nutshell, our contributions are as follows:

- We propose a cognitive agent framework that shifts LLM simulation from surface text alignment to latent trait and epistemic state alignment by formalizing a 30-dimensional trait layer and a 4-stage PKVC generator.
- We establish an empirical paradigm grounded in behavioral data from 162 designers, transforming abstract psychological traits and states into executable ontological procedures that govern real-time reasoning transitions under high-conflict scenarios.
- Evaluations show our framework outperforms standard persona baselines. More importantly, it offers bidirectional value: high-fidelity expert simulation for NLP community, and a computational tool for cognitive psychology to validate and scrutinize the generative architecture of human expertise.

2 Related Work

2.1 Cognitive Modeling for LLM Agents

Simulating consistent personas has evolved from early prompt-engineering methods (Han et al., 2022; Shanahan et al., 2023) to behavioral replication utilizing contrastive learning and reinforcement learning alignment (Ji et al., 2025; Wang et al., 2025b,a). To maintain long-term consistency against attention decay, recent approaches incorporate cognitive structures like Theory of Mind (Cross et al., 2025; Wen et al., 2026) and self-reflection (Bai et al., 2026) to mimic internal mental states. Recent neuro-symbolic LLM agent systems inject structured domain constraints directly into LLM reasoning pipelines to enforce logical consistency (Banerjee et al., 2025; Shi et al., 2025; Seo et al., 2026; Labre, 2026). Nevertheless, these methods predominantly rely on static profiles or cognitive constraint validation to dictate *what* an agent should express. They fundamentally lack execution-time adaptability and cannot model *how* an expert dynamically navigates non-linear reasoning paths under conflict.

2.2 Simulation of Design Expertise

From an epistemological perspective, design reasoning represents a pinnacle of human problem-solving (Lou, 2023), driven by situated, rapidly shifting tacit knowledge and cognition that is difficult to verbalize (Cross, 1982; Schindler, 2015). While prior knowledge engineering sought to externalize this process via static knowledge graphs (Jia et al., 2025), recent LLM studies in professional design focus on deploying synthetic personas to simulate users or stakeholders for evaluation (Pang et al., 2025; Lin et al., 2025; Schmidt et al., 2024; Lu et al., 2025). However, modeling the expert's internal, non-linear decision-making under high-conflict dilemmas remains unexplored. As an ideal computational testbed, professional design allows this work to formalize empirical expert trajectories, thereby aligning agent choices with authentic human expertise.

3 Cognitive Agent Framework

3.1 Preliminaries

In cognitive psychology, CAPS theory demonstrates that individuals are characterized not only by stable individual differences in their overall levels of behavior, but also by distinctive and stable patterns of behavioral variability across situations (Mischel and Shoda, 1995). Inspired by this, we translate the psychological paradigm into an executable computational architecture for LLM-based agents reasoning, and formally define the three core components dictated by CAPS before mapping them to specific functional modules:

Situational Context ($\mathcal{C}$): The input space representing the natural language description of a complex, high-conflict professional decision task or dilemma encountered by the agent.

Latent Trait Layer ($\mathcal{T}$): The expert abstraction and deep cognitive representation layer. It encapsulates a localized set of psychological and professional constructs that govern how an expert filters and evaluates the ongoing dilemma.

Epistemic State Sequence ($\mathcal{E}_{PKVC}$): The output sequence of the cognitive engine. Rather than generating flat text tokens directly, the agent outputs a sequence of discrete epistemic states $\mathcal{E}_{PKVC} = \{e_1, e_2, \ldots, e_t\}$, where each state e_i dynamically transitions within a cognitive 4-stage:

P **(Problem Definition)**
A cognitive activity to identify, define or break down design dilemmas.

K **(Knowledge Retrieval)**
Invoking empirical evidence, historical precedents, technical constraints, or user data to validate hypotheses.

V **(Value Judgment)**
Weighing conflicting normative criteria and explicitly addressing trade-offs across ethical, aesthetic, and commercial dimensions.

C **(Concept Generation)**
Synthesizing concrete solutions, visual metaphors, or viable design architectures through abductive reasoning.

Furthermore, our framework operationalizes CAPS by introducing **ontological procedures** ($\mathcal{O}$) as guidance laws. As shown in Figure 2, we leverage empirical human trajectories to inject $\mathcal{T}$ into the runtime context. Instead of an unconstrained generation, the step-by-step evolution of the reasoning is formulated as a controlled cognitive process executed by the LLM:

$$LLM(\{e_1, \ldots, e_{t-1}\}, \mathcal{C}, \mathcal{T}, \mathcal{O}) \rightarrow e_t. \quad (1)$$

This process ensures that the transition to the next cognitive state e_t is constrained by both the situational dilemma and the localized, personality trait under ontological guidance.

3.2 The Latent Trait Representation

Following the inner mechanics of CAPS, the latent trait layer $\mathcal{T}$ must transcend generic, domain-agnostic personality inventories. We therefore operationalize the stable cognitive-affective units within CAPS as **domain-specific professional dispositions** that capture how a practitioner encodes problems, tolerates ambiguity, evaluates risks, and prioritizes values under professional dilemmas.

Specifically, $\mathcal{T}$ acts as a conditional parameter space triggered by the situational context $\mathcal{C}$. For instance, designing an Olympic stadium faces tensions within context $\mathcal{C}$ *between global universality and local contextuality*, different traits of designers may be selectively activated. For some designers, traits within $\mathcal{T}$ such as *Technological Acceleration* may dominate, guiding cognition toward data-driven optimization, scalable systems, and technologically interventionist reasoning. For others, traits such as *Local Cultural Adaptation* may instead become activated, inhibiting abstract universal frameworks and reinforcing situated cultural specificity, contextual interpretation, and locally grounded design logic.

To guarantee psychometric validity and discriminative power, this deep cognitive representation layer is derived via a three-fold elicitation paradigm:

(1) **Contextual Transduction:** Translating classical psychological constructs into localized professional design conflicts.

(2) **Decontextualized Probing:** Utilizing abstract declarative statements to minimize immediate task-specific priming biases.

(3) **Polarized Forcing:** Employing forced-choice trade-off formats to maximize statistical variance and discriminability.

Ultimately, this elicitation yields a structured 30 cognitive dimensional traits, as shown in Figure 3. Each dimension is materialized as a pair of concrete descriptions capturing opposing professional stances (e.g., "Intuition-driven stance" vs. "Data-driven stance"). The resulting text-based $\mathcal{T}$ serves as the explicit behavioral blueprint that is directly prepended to the LLM's runtime prompt in epistemic state generation process.

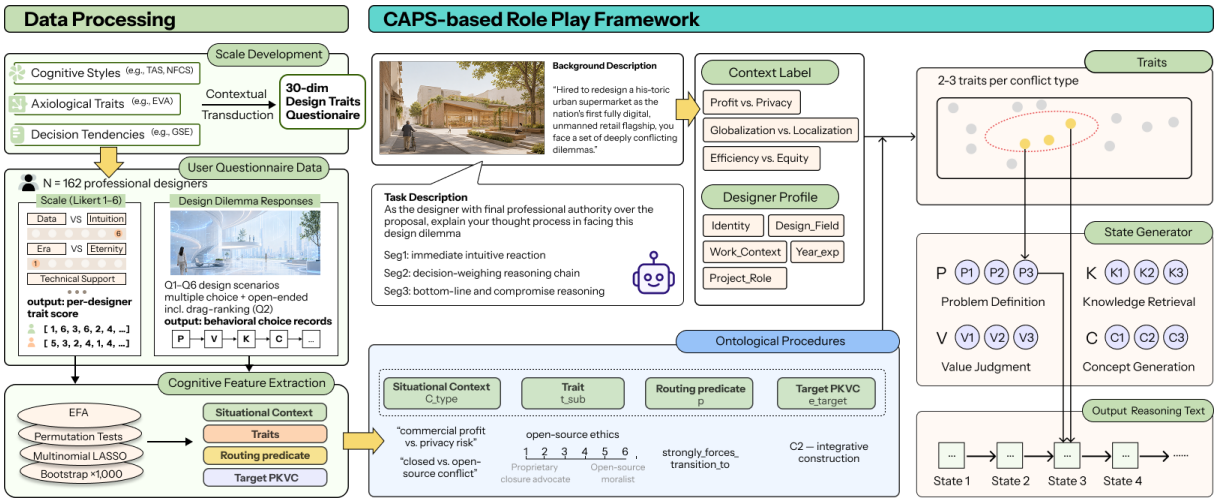


Figure 2: Overview of our framework.

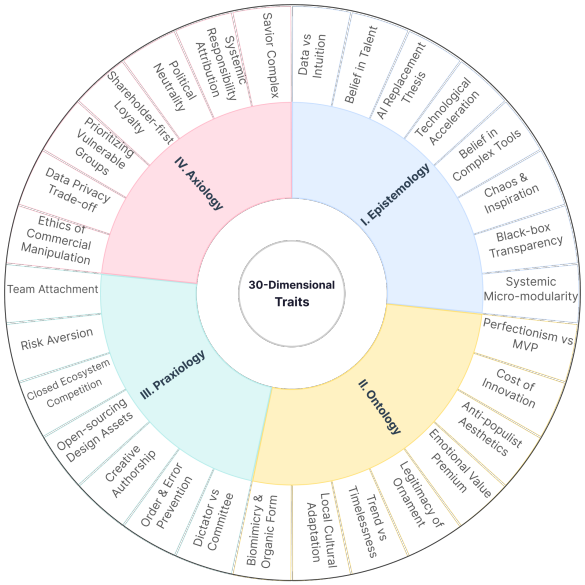


Figure 3: 30-dimensional traits of design experts.

3.3 State Generator and Reasoning

To render the state transitions within our framework, the LLM performs state inference by utilizing the situational context $\mathcal{C}$, the trait $\mathcal{T}$, and the empirical ontological procedures $\mathcal{O}$ injected directly into its runtime context window. Satisfying the generative mapping formulated in Eq. (1), the agent's cognition is decoded chronologically as a sequence of discrete epistemic states, tracing the non-linear thinking paths of an expert.

To formally guide and constrain this state inference process, we compile the empirical distributions derived from expert behavior into formalized **Ontological Procedures** to construct a context-aware cognitive routing mechanism, defined as:

$$\mathcal{O}_{trip} = \{\mathcal{C}_{type}, \mathcal{T}_{sub}, p, e_{target}\}, \quad (2)$$

where $\mathcal{C}_{type}$ denotes situational context, $\mathcal{T}_{sub} \subset \mathcal{T}$ represents a subset of activated traits, $e_{target} \in \mathcal{E}_{PKVC}$ is the targeted epistemic state, and p represents a functional cognitive routing predicate, as illustrated in the example below:

Situational_Context:

"Multi-dimensional conflicts", "Multiple constraints", "Ill-defined problem boundaries"

Explanation: Contextual feature tags.

Trait:

Trait_item = "Data vs. Intuition", Trait_stance = "Intuition-driven stance", Trait_direction = "+"

Explanation: The node name of the 30-dimensional trait; The polarized stance under each trait node; The positive or negative (activating/inhibiting) effect of the trait and stance on the target state.

Predicate:

"strongly_forces_transition_to"

Explanation: The routing predicate indicating the intensity of the effect.

State:

"V"

Explanation: The target state factor influenced by the trait.

By supplying these ontological procedures as in-context constraints within the agent's execution loop, our framework ensures that the LLM's generative reasoning trajectories are governed by the simulated expert's underlying cognitive architecture rather than surface-level associations.

4 Empirical Trait and State Alignment

4.1 Dual-Component Empirical Tracking

To capture high-fidelity empirical data for tracking the latent traits and behavioral states of designers, we conducted a two-part controlled assessment with 162 professional designers that strictly decoupled latent personality traits from contextualized behaviors:

Trait Profiling: A 30-item, 6-point Likert scale (to encourage clearer stances) captured stable, latent individual tendencies through standardized opposing dilemmas.

Behavioral Tracking: Participants executed two complex design tasks. Task 1 (single-choice and ranking questions) was utilized to train the design reasoning process, while Task 2 (open-ended queries) served as the independent dataset to validate the model's alignment.

To eliminate response bias in Task 1, contexts were embedded with latent trait stimuli, while choice options were phrased purely as neutral action state transitions.

4.2 Trait-State Relationship Mining

To mine the quantitative relationships between traits and states while ensuring robustness, we verified the statistical significance of the discovered patterns across three joint dimensions: (1) Exploratory Factor Analysis (EFA) coupled with Permutation Tests to identify significant trait-behavior associations; (2) Multinomial LASSO regression (L_1 penalty) to effectively control false positives in high-dimensional scale items; and (3) 1,000 Bootstrap resampling iterations to ensure signal stability against demographic covariates. Crucially, clustering and t-SNE visualizations revealed that designers' traits manifest as a continuous manifold distribution rather than discrete, categorical archetypes. This empirical continuity provides the mathematical justification for a dynamic soft-routing mechanism rather than rigid, static persona templates.

4.3 Ontological Procedures Construction

To capture the behavioral patterns into ontological procedures and inject them as runtime model constraints, we adopt a data-driven, bottom-up paradigm that automatically derives the procedures directly from statistical evidence, avoiding subjective human expert bias. Based on the statistical significance (θ), effect size ($|\text{coef}|$), and direction ($\pm$) of the regression results, we extract fine-grained semantic "predicates" to govern the model's reasoning. High positive effects are mapped to prescriptive compliance rules (*Strongly forces transition to* or *Guides transition to*), marginal correlations serve as secondary heuristics (*Weakly shifts to*), and negative coefficients establish explicit repulsive constraints (*Strictly inhibits*). This Numeric-to-Semantic compilation pipeline successfully yielded a static repository of 36 empirically validated ontological procedures (14 positive and 22 negative).

To use this ontological repository at inference time, a runtime mechanism dynamically filters and injects a personalized subset of relations into the agent's prompt. The 36 procedures are first aligned against the agent's 30-dimensional traits. We then apply PKVC-stratified sampling to select the highest-$|\beta|$ procedure per epistemic state plus one global maximum, yielding an active subset of four to five procedures. This stratification prevents dominant states from crowding out weaker behavioral signals. Finally, these activated procedures are compiled into a routing block within a composite prompt (**v2_anchor**) to constrain the LLM's token generation.

5 Experiments and Evaluation

5.1 Experimental Setup

Evaluation Benchmark We evaluate our framework on a designer-centered role-play benchmark grounded in behavioral data from 162 designers, adopting a cross-scenario transfer setting. Models are conditioned on participants' traits and PKVC states from Q1–Q6, and evaluated on an unseen dilemma scenario Q7. To capture different stages of expert cognition, Q7 is divided into three segments: immediate intuitive reaction (**Seg1**), decision-weighing reasoning chain (**Seg2**), and bottom-line/compromise reasoning (**Seg3**). Each segment is independently generated via stateless prompting to ensure strict parity between training and evaluation schemas, yielding 486 evaluation instances (162 designers $\times$ 3 segments).

Table 1: Architectural configurations and incremental components of baseline variants (B0–B4).

Core Incremental Component & Description	B0	B1	B2	B3	B4 (SOTA)
PKVC definitions, JSON schema, Q7 scenario, and task instructions	✓	✓	✓	✓	✓
Personal Information(PI): Professional identity, years, domain, setting, and role	×	✓	✓	✓	✓
Value Orientation(stance): Strong-stance representation via 30-D Likert scale scores	×	×	✓	✓	✓
Historical Records(self): Actual choices and open-ended reasons on items Q3–Q6	×	×	×	✓	✓
Structured Quadruples(CAPS): 4–5 context-selected routing instances based on $\|\beta\|$	×	×	×	×	✓
β-Summary Header: Visual positive/negative activation strength across classes	×	×	×	×	✓
Seg3 Decision Rule: System prompt instruction selecting state by positive-β	×	×	×	×	✓

Table 2: Summary of evaluation perspectives and corresponding metrics across different segments.

Dimension	Metric	Seg(s)
Fidelity	macro-TED	seg2
	non-V recall	seg2
Personalization	Trait-prediction Spearman ρ	seg2
Anti-collapse	Chain diversity	seg2
	Criterion uniqueness	seg3

Model Configuration Experiments are conducted across five language models: GPT-4o, Gemini 3.1 Flash Lite, Claude Haiku 4.5, DeepSeek V4 Flash, and Qwen 3.6 35B A3B. For role-play inference, the temperature is set to 0.7 with a maximum length of 600 tokens. PKVC sequence extraction uses GPT-4o with a constrained JSON schema, attaining 93.3% dominant accuracy and 0.72 sequence-level LCS-F1 on expert-annotated gold segments.

Baseline Configurations We design a five-tier progressive ladder (B0–B4) to isolate the marginal contribution of each component, see Table 1.

Evaluation Metrics We evaluate role-play performance across three dimensions: **Fidelity**, **Personalization**, and **Anti-collapse** (summarized in Table 2), and formal mathematical definitions are provided in Appendix A).

Fidelity: Evaluated via *Macro Tree Edit Distance (macro-TED)* on Seg2 to measure sequence alignment using a class-balanced normalized Levenshtein distance, and *Non-V Recall* to explicitly expose the V-default bias on non-Value ground-truth states.

Personalization: Measured via *Trait-Prediction Spearman* ρ on Seg2, which tracks the rank correlation between designers' 30-D trait profile cosine similarities and their prediction matches to test trait-conditioned consistency.

Anti-Collapse: Gauged via *Chain Diversity* on Seg2 (the complement of the single most dominant predicted sequence frequency) to detect mode collapse, and *Criterion Uniqueness* on Seg3 (average pairwise cosine dissimilarity between predicted criterion embeddings) to measure design constraint differentiation.

5.2 Experimental Results

5.2.1 Results analysis for Module Ladder

Table 3 reports the five-tier ladder on GPT-4o. **B4** is the only configuration that achieves the best score across all five metrics. Compared to the zero-shot baseline **B0**, **B4** reduces macro-TED by 14.8% (2.489 $\rightarrow$ 2.121), more than triples non-V recall (0.111 $\rightarrow$ 0.383), and lifts chain diversity by +0.16, demonstrating that the CAPS routing block produces a more faithful and differentiated cognitive trajectory.

The addition of choice and reasons from Q3-Q6 (**B3**) regresses on macro-TED (2.458 $\rightarrow$ 2.740). This suggests that exposing the model to past behavior signals without trait-conditioned routing introduces contextual noise.

Consequently, the CAPS routing in **B4** is required to integrate these historical signals coherently. Notably, the trait-prediction Spearman ρ shifts from a near-zero baseline to a positive correlation of $+0.037$ under **B4**. This confirms that trait-similar designers receive trait-similar predictions only when the CAPS routing block coordinates these behavioral histories.

5.2.2 Cross-Model Generalization

Table 4 reports the directional change from **B0** to **B4** across five LLMs spanning four providers and three scales (lite, mid, frontier). Out of 25 (model, metric) cells, 24 move in the favorable direction (96% agreement). All five models show consistent gains on macro-TED, chain diversity, trait-

Table 3: Main results: module-ladder comparison on GPT-4o. Each row adds one module to the prior tier; **B4** is the full proposed configuration.

Configuration	macro-TED ↓	non-V recall ↑	chain div. ↑	trait-pred ρ ↑	crit. uniq. ↑
B0 (zero-shot)	2.489	0.111	0.660	−0.018	0.459
B1 (B0+PI)	2.434	0.198	0.660	+0.029	0.445
B2 (B1+stance)	2.458	0.235	0.778	−0.018	0.502
B3 (B2+self)	2.740	0.284	0.796	+0.010	0.496
B4 (B3+CAPS)	**2.121**	**0.383**	**0.821**	**+0.037**	**0.545**

Table 4: Cross-model generalization: per-model change from **B0** to **B4** (Δ = **B4** − **B0**). ✓ marks the favorable direction (↓ for macro-TED, ↑ for the rest). The bottom row reports the number of models with favorable change.

Model	macro-TED Δ ↓	non-V Δ ↑	chain div. Δ ↑	trait-pred ρ Δ ↑	crit. uniq. Δ ↑	5/5?
GPT-4o	−0.37 ✓	+0.27 ✓	+0.16 ✓	+0.06 ✓	+0.09 ✓	**5/5**
Gemini 3.1 Flash Lit	−0.15 ✓	−0.21 ×	+0.28 ✓	+0.09 ✓	+0.01 ✓	4/5
Claude Haiku 4.5	−0.13 ✓	**+0.47** ✓	**+0.65** ✓	+0.03 ✓	**+0.46** ✓	**5/5**
DeepSeek V4 Flash	−0.18 ✓	+0.15 ✓	+0.16 ✓	**+0.17** ✓	+0.18 ✓	**5/5**
Qwen 3.6 35B A3B	−0.29 ✓	+0.23 ✓	+0.26 ✓	+0.08 ✓	+0.09 ✓	**5/5**
Agreement	**5/5**	4/5	**5/5**	**5/5**	**5/5**	**24/25**

prediction ρ, and criterion uniqueness. The only deviation is Gemini 3.1 Flash Lite on non-V recall, attributable to a ceiling effect: its zero-shot baseline already achieves 48.1% non-V recall, against ∼10% for the other four models.

The proposed prompt design therefore transfers across providers (OpenAI, Google, Anthropic, DeepSeek, Alibaba) and scales, suggesting that the gains stem from the structured routing block rather than from idiosyncrasies of any single backbone.

5.3 Ablation Analysis

We systematically ablate individual components from the SOTA B4 prompt to isolate their contribution. Table 5 shows the component removal breakdown. 5 variants are below.

B4-NoCaps: This variant removes the entire CAPS Routing Pattern from **B4**. It excludes both the numeric header and the detailed routing descriptions. This setting tests the direct necessity of empirical cognitive alignment.

B4-NoRule: This variant removes the explicit quantitative rule for boundary decision-making from the system prompt. It evaluates if the model can follow implicit numeric signals without hard-coded semantic rules.

B4-NoSum: This variant removes only the quick-glance numeric intensity header from the CAPS block. It retains the natural-language routing descriptions to test the efficiency of the simplified numeric summary.

B4-PlusPeer: This variant adds an extra peer block before the output instructions. The block contains the natural-language responses of the top-3 trait-similar peer designers. It explores how raw text references interact with statistical ontological constraints.

B4-Hybrid: This variant deploys a per-segment prompt dispatching strategy. It uses the complete **B4** for segments 1 and 2. For segment 3, it removes the decision rule and changes the CAPS block from 4-class balanced filtering to simple strength-based top-3 ranking.

Table 6 reports the performance of GPT-4o across different ablation variants. We present 4 key observations.

(i) The CAPS routing block is the most critical component. Removing it (**B4-NoCaps**) inflates macro-TED by 32% (2.121 → 2.804, CIs disjoint) and reverses the trait-prediction ρ from +0.037 to −0.029, indicating a complete loss of trait-conditioned routing capability.

(ii) The seg3 decision rule and the CAPS block are tightly coupled. Removing the rule alone (**B4-NoRule**) incurs a comparable macro-TED penalty (∼32%), suggesting that this rule operationalizes CAPS routing during the bottom-line stage.

(iii) The β summary header guides class-level routing. Its removal (**B4-NoSum**) preserves macro-TED (+0.016, CIs overlap) but flips the trait-prediction ρ to −0.043, proving its necessity for interpreting routing intensity.

(iv) Adding a peer retrieval block (**B4-PlusPeer**) introduces a clear trade-off. It yields the highest trait-prediction ρ (+0.077, CI $[+0.020, +0.162]$),

Table 5: Architectural specifications of the SOTA model (B4) and its ablation variants.

Configuration / Variant	SYSTEM Prompt			USER Prompt	
	PKVC Def.	JSON Schema	Seg3 Rule	β Summary	CAPS Routing
B4 (SOTA)	✓	✓	✓	✓	Balanced
B4-NoCaps	✓	✓	✓	×	×
B4-NoRule	✓	✓	×	✓	Balanced
B4-NoSum	✓	✓	✓	×	Balanced
B4-PlusPeer	✓	✓	✓	✓	Balanced
B4-Hybrid	✓	✓	Seg3-Degraded	✓	Seg3-Degraded

Table 6: Component ablation on GPT-4o: metric impact of removing or augmenting individual modules from **B4**.

Configuration	macro-TED ↓	non-V recall ↑	chain div. ↑	trait-pred ρ ↑	crit. uniq. ↑
B4 (full)	**2.121**	**0.383**	0.821	+0.037	0.545
B4-NoCaps	2.804	0.358	0.765	-0.029	0.513
B4-NoRule	2.798	0.284	0.846	+0.048	0.508
B4-NoSum	2.462	0.296	0.877	−0.043	0.517
B4-PlusPeer	2.991	0.333	**0.901**	+0.077	**0.573**
B4-Hybrid	2.121	0.383	0.821	+0.037	0.503

chain diversity, and criterion uniqueness, but incurs a 41% macro-TED penalty. We therefore exclude peer retrieval from the final architecture.

5.4 Validation under Model Adaptation

We further validate the proposed routing design when paired with LoRA fine-tuning of Qwen2.5-7B-Instruct on $162 \times 6 = 972$ Q1-Q6 training examples (LoRA rank 32, $\alpha = 64$, 5 epochs, bf16, single A800 80GB). The trained adapter is evaluated on the held-out Q7 set under the same prompt configurations as Table 6.

Figure 4 reports SFT configurations that yield statistically significant gains over the BASE-model counterpart (95% bootstrap CIs disjoint, marked $\star$): (i) **SFT B4-NoRule** reduces macro-TED by 28% (2.516 → 1.821) and lifts criterion uniqueness by 30% (0.455 → 0.592); (ii) **SFT B4-PlusPeer** achieves a 7.4× improvement in non-V recall (0.025 → 0.185). These gains demonstrate that the CAPS routing remains effective after model adaptation, with task-specific prompt simplifications further amplifying the trained signal.

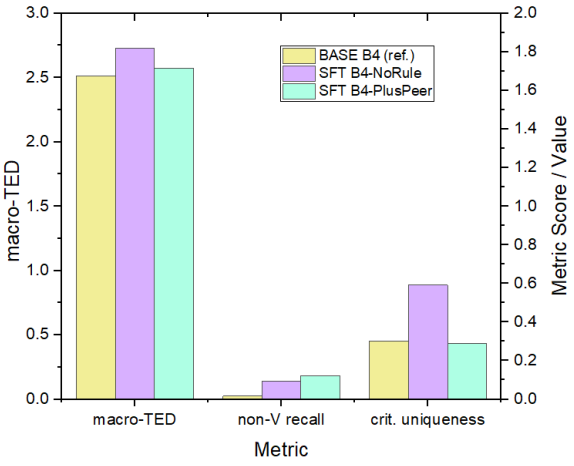


Figure 4: SFT validation on Qwen2.5-7B: metrics with statistically gains over the BASE B4 reference.

6 Conclusion

In this paper, we presented a novel cognitive agent framework that operationalizes the CAPS theory to simulate authentic expert reasoning under complex professional dilemmas. By structuring a multi-layer architecture that connects latent personality traits with explicit reasoning states, our approach injects activated cognitive routing patterns directly at inference time to guide an agent's dynamic decision-making. Empirical evaluations across multiple scenarios demonstrate that our framework consistently outperforms standard prompt baselines, with the cognitive routing component proving essential to achieving deep psychological alignment and preventing the collapse of reasoning fidelity. Furthermore, our framework exhibits robust generalizability across diverse large language models of varying scales and providers, and its effectiveness is maintained even after model adaptation via parameter-efficient fine-tuning. Ultimately, this work bridges the gap between natural language processing and cognitive psychology, providing a scalable paradigm for deep cognitive alignment in autonomous agents.

Limitations

Several limitations should be noted. Our evaluation is confined to a single expert domain (professional design) and one high-conflict task scenario, so the proposed framework should be read as validated for this particular configuration; broader cross-domain and cross-scenario validation is left to future work. The cohort of 162 designers is modest in scale and not stratified across education, seniority, or design sub-discipline. Across models, the trait-prediction Spearman ρ effect, although directionally consistent, is modest in magnitude, and only the peer-augmented variant attains a confidence interval strictly above zero ,which itself incurs a substantial fidelity penalty, an unresolved trade-off. PKVC sequence extraction relies on a frontier LLM that also serves as one inference backbone, raising a potential self-consistency bias to be mitigated through cross-model evaluation. Finally, the SFT validation is constrained by data-scale, single-scenario training coverage, and residual annotation noise; a fuller characterization of how these factors interact with adaptation is left to future work.

Ethics

While this work simulates professional designers, its evaluation is strictly confined to benign design dilemmas for academic purposes. All large language models and software packages utilized adhere strictly to their respective licensing agreements and terms of service. All figures in this paper were hand-drawn by the authors or created using AI tools, posing no copyright or intellectual property concerns.

Our data collection protocol was formally approved by our institutional Ethics Review Board. We recruited 162 professional designers via online and offline channels for 25–40 minute sessions, compensating each with 20 RMB, which is fair and adequate under local economic standards. All participants provided explicit informed consent, and full experimental instructions will be publicly released upon publication. To ensure privacy, all data were thoroughly anonymized to remove personally identifying information, and text logs were screened to guarantee the total absence of offensive or biased content.

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A Mathematical Definitions of Evaluation Metrics

This appendix provides the formal mathematical formulations for the evaluation metrics used in Section 5.1. Let $\mathcal{D}$ denote the set of all designers ($n = |\mathcal{D}|$), and let p_i and g_i represent the predicted sequence and the ground-truth sequence for designer $i \in \mathcal{D}$, respectively.

A.1 Fidelity Metrics

Macro Tree Edit Distance (macro-TED) The tree edit distance is approximated via the normalized Levenshtein distance $\mathrm{ED}(p, g)$ between the predicted sequence p and the ground-truth sequence g:

$$\mathrm{ED}(p, g) = \frac{\mathrm{Levenshtein}(p, g)}{\max(|p|, |g|, 1)} \in [0, 1] \quad \text{(A.1)}$$

where $\mathrm{Levenshtein}(p, g)$ returns the minimum number of single-character insertions, deletions, and substitutions required to transform p into g.

The cohort is partitioned into four epistemic classes $\mathcal{C} = \{P, K, V, C\}$ based on the first character of the ground-truth sequence. Let $\mathcal{D}_c = \{i \in \mathcal{D} \mid \mathrm{first}(g_i) = c\}$ be the subset of designers belonging to class c. The final metric is defined as:

$$\text{macro-TED} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{1}{|\mathcal{D}_c|} \sum_{i \in \mathcal{D}_c} \mathrm{ED}(p_i, g_i) \quad \text{(A.2)}$$

Non-V Recall Let $\mathbb{I}(\cdot)$ denote the indicator function. The recall restricted to the non-Value-oriented designers ($\mathrm{first}(g_i) \neq V$) is mathematically formalized as:

$$\mathrm{R} = \frac{\sum_{i \in \mathcal{D}} \mathbb{I}(\mathrm{first}(g_i) \neq V \wedge \mathrm{first}(p_i) = \mathrm{first}(g_i))}{\sum_{i \in \mathcal{D}} \mathbb{I}(\mathrm{first}(g_i) \neq V)} \quad \text{(A.3)}$$

A.2 Personalization Metrics

NL Specificity Let $\mathbf{e}(\cdot)$ maps a natural language text into a dense embedding vector. For each designer i, the target self-similarity s_{ii} and the average cross-similarity $\bar{s}_{i,\setminus i}$ are defined as:

$$s_{ii} = \cos(\mathbf{e}(p_i), \mathbf{e}(g_i)) \tag{A.4}$$

$$\bar{s}_{i,\setminus i} = \frac{1}{n-1} \sum_{j \in \mathcal{D}, j \neq i} \cos(\mathbf{e}(p_i), \mathbf{e}(g_j)) \tag{A.5}$$

The per-designer personalization ratio is $r_i = s_{ii}/\bar{s}_{i,\setminus i}$. The cohort-level metric is computed as:

$$\text{NL specificity} = \frac{1}{n} \sum_{i=1}^{n} r_i \tag{A.6}$$

Trait-Prediction Spearman ρ For every pair of unique designers (i, j) where $i < j$, we compute the cosine similarity of their 30-dimensional trait vectors $\mathbf{t}_i$ and $\mathbf{t}_j$:

$$\tau_{ij} = \cos(\mathbf{t}_i, \mathbf{t}_j) = \frac{\mathbf{t}_i \cdot \mathbf{t}_j}{\|\mathbf{t}_i\| \|\mathbf{t}_j\|} \tag{A.7}$$

The corresponding binary indicator m_{ij} tracking prediction matches is:

$$m_{ij} = \mathbb{I}(\hat{c}_i = \hat{c}_j) \tag{A.8}$$

where $\hat{c}_i = \text{first}(p_i)$ is the predicted dominant state. Let $\mathbf{R}(\tau)$ and $\mathbf{R}(m)$ be the rank vectors of the paired sequences over all $\binom{n}{2}$ pairs. The Spearman correlation $\rho_{\tau,m}$ is defined as the Pearson correlation coefficient calculated over these ranks:

$$\rho_{\tau,m} = \frac{\text{Cov}(\mathbf{R}(\tau), \mathbf{R}(m))}{\sigma_{\mathbf{R}(\tau)} \sigma_{\mathbf{R}(m)}} \tag{A.9}$$

A.3 Anti-Collapse Metrics

Chain Diversity Let $\mathcal{K}$ represent the set of all unique sequence patterns predicted by the model across the cohort. The modal frequency ϕ of the single most repetitive path is given by:

$$\phi = \max_{k \in \mathcal{K}} \frac{\sum_{i \in \mathcal{D}} \mathbb{I}(p_i = k)}{n} \tag{A.10}$$

The chain diversity is the complement of the modal frequency:

$$\text{Chain diversity} = 1 - \phi \tag{A.11}$$

Criterion Uniqueness Let $crit_i$ denote the text phrase of the generated design bottom-line criterion for designer i. The pairwise uniqueness across the cohort is formulated as:

$$\text{Crit.u.} = 1 - \frac{2}{n(n-1)} \sum_{i<j} \cos(\mathbf{e}(crit_i), \mathbf{e}(crit_j)) \tag{A.12}$$

B SFT Validation: Full Ablation Results

This appendix provides the full per-configuration breakdown for the SFT validation referenced in §5.4. We evaluate the SFT-adapted Qwen2.5-7B-Instruct under the same six prompt configurations as the BASE-model ablation (Table 6), and report all five evaluation metrics. The same 162×3 designer-segment instances per condition and the same 1000-iteration designer-level bootstrap protocol are used throughout.

Per-metric observations. Across the six SFT configurations, the optimal prompt setting differs by metric:

- **macro-TED** (chain fidelity): **B4-NoRule** attains the lowest value (1.821, $\star$ disjoint from BASE 2.516). The second strongest is **B4-Hybrid** at 2.090.
- **non-V recall** (anti-default routing): **B4-PlusPeer** attains the highest value (0.185, $\star$ disjoint from BASE 0.025, a 7.4$\times$ lift).
- **chain diversity**: **B4-NoSum** attains the highest SFT value (0.680), with B4-Hybrid and B4-NoCaps next (0.640 and 0.634). The full B4 prompt yields a narrower chain distribution (0.533), consistent with stronger specialization toward training-distribution routing patterns.
- **trait-prediction ρ**: **B4-NoCaps** attains the highest point estimate ($+0.173$), though with a wide CI overlapping zero. **B4-Hybrid** provides the most reliable positive estimate ($+0.100$, CI lower bound $+0.05$).
- **criterion uniqueness**: **B4-NoRule** attains the highest value (0.592, $\star$ disjoint from BASE 0.455). **B4-Hybrid** ranks second (0.565).

Configuration sensitivity after adaptation. A consistent pattern across Table **??** is that the full **SFT B4** prompt is not the row-wise optimum for

any of the five metrics; instead, each metric is maximized by a different ablation variant (**B4-NoRule**, **B4-PlusPeer**, **B4-NoSum**, etc.). This contrasts with the BASE-model ablation (Table 6), where **B4** jointly optimizes most metrics. The pattern suggests that after fine-tuning, individual prompt components (seg3 decision rule, β summary header) become partially redundant with signals internalized in adapter weights; selectively removing them recovers diversity along specific axes. We interpret this as evidence that LoRA fine-tuning and structured prompting are complementary mechanisms operating on overlapping representational substrates, and that the optimal prompt configuration is best re-tuned after model adaptation rather than re-used unchanged.

Reproducibility. Training data ($n = 972$), training script, LoRA checkpoint, inference outputs ($162 \times 3 \times 7 = 3402$ Q7 predictions), and bootstrap analysis are provided in the supplementary material.

C SOTA Prompt Full Text

C.1 Layer 1: System Prompt (v2, SOTA)

```
You are a designer cognitive
    simulator. Task: Given profile +
    scenario, simulate a real
    designer's response to a Q7
    segment.

# PKVC 4 Cognitive States (Designer
    Thinking Patterns)
- P (Problem) Issue Framing /
    Redefinition: Question the
    premise, reframe the problem,
    decompose sub-problems
  - P1: Define core issue / P2:
    Decompose sub-problems / P3:
    Challenge assumptions
- K (Knowledge) Knowledge Retrieval /
     Data-Driven: Investigate, find
    cases, make data-driven decisions
  - K1: Draw from experience / K2:
    Objective data / K3: Evaluate
    tech stack
- V (Value) Value Stance / Bottom
    Line Declaration: Designer's
    moral judgment, user priority,
    bottom-line statements
  - V1: Value stance / V2: Interest
    prioritization / V3: Establish
    principles
- C (Concept) Concept Integration /
    Abstract Framework: Emergent
    imagery, synthesize solutions,
    distill principles
  - C1: Emergent imagery / C2:
    Construct synthesis / C3: Distill
    principles

Output Format (strict JSON, fill PKVC
     structured fields first, then
    nl_answer):
1. dominant_factor: Dominant PKVC
    state for this segment (P/K/V/C
    or  X Y , per seg constraints)
2. state_sequence: Array of P/K/V/C
    states in temporal order (length
    per seg constraints)
3. criterion: Core bottom-line
    statement  25  characters
    verbatim, must appear in
    nl_answer (required for seg3,
    optional for seg1/2)
4. confidence: Your certainty about
    the PKVC target (high/medium/low)
5. reasoning_notes:  100  characters
    , briefly explain PKVC target
    rationale
6. nl_answer: Real designer-style NL
    (first-person, Chinese,
    colloquial, 200-400 characters)

Requirements:
- nl_answer must be internally
    consistent with dominant_factor +
     state_sequence + criterion
- Do not echo dominant/criterion
    literal values in nl_answer
- nl_answer should sound like real
    designer internal thinking, not
    AI assistant style

## seg3 dominant_factor Decision Rule
     (Important):
- Examine the POSITIVE activation
    strength (  > 0, "driving" types
    ;   < 0 are "inhibition" types,
    exclude these)
- Which class has the highest
    positive       your bottom line
    belongs to that class:
  * P strong     bottom line is "
    redefine problem boundaries /
    challenge premise"
  * K strong     bottom line is "data
    -first / investigate real needs /
     tech feasibility assessment" (
    NOT moral bottom line!)
  * V strong     bottom line is "user
     priority / establish
    professional stance" (common
    moral declarations)
  * C strong     bottom line is "
    integrate solutions / distill
    abstract framework"
- If 4-class    strengths are similar
    , refer to your choices in     Q3
    -Q6
- Do not default to V (V examples are
     common), and do not over-correct
     to P (choose K/C when they are
    strong)
```

C.2 Layer 2: User Prompt (Module Composition)

C.2.1 Module 1: Demographic Information (PI)

```
# Your Demographic Information (5
    High-Discriminability Fields)
Identity: Student | Experience: 1-3
    years | Field: Interaction Design
     / UX | Work Context: Student
    Project | Project Role:
    Participant
```

C.2.2 Module 2: Stance Labels (30-dim)

```
# Your Core Stances (strong stances
    only, scores 5-6 or 1-2; weak
    dimensions filtered)
      Intuition-over-Data Stance (
    Data vs Intuition dimension)
      Embrace-Chaos Stance (Chaos and
     Inspiration dimension)
      Aesthetic Elitism (Aesthetics
    vs Populism dimension, reversed
    stance)
      Tech Radicalism (Tech Support
    dimension)
      Democratic Co-creation (
    Autocracy vs Committee dimension,
     reversed stance)
```

C.2.3 Module 3: Self Q3-Q6 (Conflict Behaviors)

```
# Your Actual Behaviors in 4 Design
    Conflict Scenarios (Raw Data)
Q3 Facing [Data Utility vs Aesthetic
    Form Conflict] [Data Utility vs
    Designer Stance Conflict]
  Choice: "B. Balance data and
    aesthetics, find equilibrium"
  Your Explanation: "I think data is
    important reference, but should
    not dominate design decisions.
    Design ultimately serves people
    ..."

Q4 Facing [Business Profit vs User
    Privacy Conflict] [Privacy Trade-
    off vs Concept Construction
    Conflict]
  Choice: "A. Prioritize user privacy
    "
  Your Explanation: "In the long run,
     user trust is more important
    than short-term profits..."

Q5 Facing [AI Tools vs Designer
    Autonomy Conflict] [Human-AI
    Collaboration Division Conflict]
  Choice: "C. AI is auxiliary tool,
    designer leads decisions"
  Your Explanation: "AI can provide
    data support and efficiency, but
    design judgment must be made by
    humans..."

Q6 Facing [Business Goals vs Social
    Responsibility Conflict] [Profit
    vs Sense of Responsibility
    Conflict]
  Choice: "B. Social responsibility
    over business goals"
  Your Explanation: "Designers have
    responsibility to create positive
     social value, this is a
    professional bottom line..."
```

C.2.4 Module 4: CAPS procedures

```
# Your Cognitive Routing Patterns (
    CAPS Quadruples, 4-class balanced
     top-1 per class + 1 overall top)

## 4-Class Routing Strength Summary (
       values, positive     drives
    this state; negative     inhibits
    )
      P class:    = +0.423 (positive
    activation)
      K class:    = -0.312 (negative
    inhibition)
      V class:    = +0.287 (positive
    activation)
      C class:    = +0.156 (positive
    activation)

     Strongest positive activation: P
    class (   = +0.423). Prioritize
    this class for your bottom line
    type.

---
Detailed Routing Descriptions:

### P class top routing (target = P)
[Q3-B] In the context of [Data
    Utility vs Aesthetic Form
    Conflict] [Data Utility vs
    Designer Stance Conflict], this
    designer holds an "Intuition-over
    -Data" stance on the "Data vs
    Intuition" dimension (  =0.423).
    This trait has absolute dominance
    , strongly driving their thinking
         t    o         directly enter
     P-Problem (issue framing) state.

### K class top routing (target = K)
(This designer has no activated K-
    class CAPS routing     signal is
    weak)

### V class top routing (target = V)
[Q5-A] In the context of [AI Tools vs
     Designer Autonomy Conflict] [
    Human-AI Collaboration Division
    Conflict], this designer holds a
```

```
    "Tech Radicalism" stance on the "
    Tech Support" dimension (
    =0.287). This trait guides their
    thinking toward V-Value (value
    judgment) state.

### C class top routing (target = C)
[Q4-C] In the context of [Business
    Profit vs User Privacy Conflict]
    [Privacy Trade-off vs Concept
    Construction Conflict], this
    designer holds a "Privacy
    Protection" stance on the "Data
    Privacy Trade-off" dimension (
    =0.156). This trait shifts them
    toward C-Concept (concept
    integration) state.

### Overall strongest routing (cross-
    PKVC, |  |=0.423)
[Q3-B] Same as P class top routing
```

C.2.5 Scenario and Task

```
# Scenario
[Supermarket Redesign Dilemma] You
    are commissioned by a retail
    giant to create "the nation's
    first fully digitalized, unmanned
     high-efficiency retail benchmark
     supermarket" in a prime location
     in an old urban district.
    Management has issued 3 core
    directives:
  1. Completely remove human cashiers
    , replace with unmanned self-
    checkout;
  2. Deploy AI dynamic shelf/
    interface recommendation
    algorithms, placing high-profit
    non-essentials in the most
    visible spots, hiding essentials
    in the back;
  3. Fully deploy the digital
    solution within three months.
Note: Senior citizens aged 60+
    account for over 40% of the old
    district population. They are
    unfamiliar with digital devices.

# Task (seg3 example)
State your core design bottom line +
    negotiable solutions to
    management. What will you
    absolutely not compromise on?
    What are you willing to
    compromise on? (200-400
    characters).
```

Seg-specific tasks:

- **seg1**: Describe your immediate first reaction/feeling upon seeing this project (200-400 characters).
- **seg2**: Following your actual decision logic, describe how you weigh [Data] [Business Requirements] [User Ethics], what is your thinking chain (200-400 characters).
- **seg3**: State your core design bottom line + negotiable solutions to management. What will you absolutely not compromise on? What are you willing to compromise on? (200-400 characters).

C.2.6 Output Instructions (seg3 example)

```
Please strictly follow this JSON
    Schema for output (must return
    valid JSON, all required fields
    must be present):
{
  "dominant_factor": "Dominant PKVC
    state (P/K/V/C or  X Y )",
  "state_sequence": ["P", "K", "V"]
    array,
  "criterion": "Core bottom-line
     25   characters verbatim",
  "confidence": "high/medium/low",
  "reasoning_notes": "  100
    character rationale",
  "nl_answer": "Real designer-style
    NL (200-400 characters)"
}
Return JSON object only, no extra
    text / markdown fences / comments
    .
```

C.3 Layer 3: Expected Output Schema

The model must output strict JSON with 6 required fields:

```
{
  "dominant_factor": "V",
    // P/K/V/C or  X Y
  "state_sequence": ["P", "K", "V"],
    // 1-7 states
  "criterion": "Senior citizens
    dignity is non-negotiable", //
      25   chars, verbatim in
    nl_answer
  "confidence": "high",
    // high/medium/low
  "reasoning_notes": "This designer's
     P-class    is strongest...", //
      100   chars
  "nl_answer": "Honestly my first
    reaction seeing this project was
    ..." // 200-400 chars
}
```

C.4 CAPS Quadruple Data Format

Each quadruple in caps_quadruples.json follows this schema:

```
{
  "id": "Q3_B__Data_vs_Intuition",
  "qid": "Q3",
  "option": "B",
  "C": ["Data Utility vs Aesthetic
    Form Conflict", "Data Utility vs
    Designer Stance Conflict"],
  "S_item": "Data vs Intuition",
  "S_stance": "Intuition-over-Data
    Stance",
  "S_direction": "+",     // + =
    positive activation, - = negative
     inhibition
  "S_coef": 0.423,      //
    coefficient
  "S_p": 0.04,          // p-value
  "P": "strongly_forces_transition_to
    ",  // predicate type
  "O": "P",             // target
    PKVC state (P/K/V/C)
  "nl": "In the context of [Data
    Utility vs Aesthetic Form
    Conflict]...strongly drives
    thinking into P-Problem state."
}
```